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RESEARCH ARTICLE

Human Pressures Drive Compositional Homogenization and Functional Shifts in Southeast Asian Mammal Communities

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ABSTRACT

Human activities strongly alter wildlife distributions and behaviors, rapidly reshaping their community structure. Although links between human pressures and community assembly have been explored, practical integration of multiple biodiversity indicators, especially compositional homogenization and functional trait patterns, remains limited. We addressed this gap by jointly assessing the compositional and functional responses of mammal communities to human pressures across Southeast Asia. We compiled an extensive camera-trapping dataset from 4508 sites spanning > 38° of latitude across tropical and temperate forests in Southeast Asia. We quantified compositional homogenization and functional diversity trait patterns in relation to human pressures within and across forests. Our results show that community assembly is broadly affected by human pressures, with communities in Tropical & Subtropical Moist Broadleaf Forests (TSMBF) exhibiting greater vulnerability than those in Temperate Coniferous Forests (TCF), as evidenced by pronounced compositional homogenization and distinct changes in functional traits. High-intensity human modification promoted compositional homogenization alongside functional trait divergence, while long-term changes in building-density drove functional trait convergence. This study demonstrates that integrating compositional and functional indicators provides a more holistic understanding of anthropogenic impacts on biodiversity. The heightened vulnerability of tropical mammal communities underscores the urgent need for targeted conservation strategies in these regions to curb the replacement of endemic specialists with widespread generalists.

1 | Introduction

Anthropogenic disturbances have become significant direct and indirect stressors for wildlife (Cross et al. 2021). Land-use change, urban expansion, and infrastructure development are reshaping landscapes and ecosystem functions across terrestrial, coastal, and island environments, fragmenting once-continuous habitats into isolated, scarce, or unsuitable spaces (Cloern et al. 2016; Moustakas et al. 2025; Watson et al. 2016). In addition to habitat change, human activities also alter wildlife behavior and distributions through fear landscapes, exploitation, grazing, and related pressures (Cravino et al. 2024; Suraci et al. 2019; Symes et al. 2018). Collectively, these pressures can modify community assembly and biodiversity patterns, shifting both species composition and functional diversity among communities (Suraci et al. 2019).

Despite substantial research advances, the complexity and context-dependence of human–biodiversity interactions across space constrain broad generalizations. Research on

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human–biodiversity interactions also remains uneven across taxa and regions: many studies are localized, fewer span regional to broader scales, and comparative analyses across biomes remain scarce (Cantera et al. 2023; Iijima et al. 2023; Keck et al. 2025; Stewart et al. 2019). Work in temperate and Neotropical regions far exceeds that in tropical Asia and Africa, despite the latter's exceptional biodiversity and escalating threats (Matuoka et al. 2020; Meyer et al. 2015; Resende et al. 2025). Taxonomically, mammals are underrepresented relative to research on plants, invertebrates, fishes, and birds (Cantera et al. 2023; Keck et al. 2025; Matuoka et al. 2020; Zambrano et al. 2019), even though they are particularly susceptible to human pressures, which are expected to intensify (Baisero et al. 2020; Wan et al. 2019). Achieving the goals of the Kunming–Montreal Global Biodiversity Framework to reverse biodiversity loss and enhance ecosystem integrity (Stephens 2023) requires complementary evidence on how human pressures influence community assembly and how species responses may vary by stressor type and by geographic and climatic context.

Studies often emphasize a single indicator—either compositional homogenization (Blowes et al. 2024; Keck et al. 2025; Olden and Poff 2004; Peng et al. 2025) or within-community functional diversity (Côte et al. 2023; Li, Yang, et al. 2022; Li, Hu, et al. 2022; Matuoka et al. 2020; Resende et al. 2025). These indicators are not interchangeable. Compositional homogenization, on the basis of species identities, reflects shifts in the species pool and can arise from defaunation, substitution by generalists, or stochastic drift (Coronel-Arellano et al. 2018; Li, Yang, et al. 2022; Li, Hu, et al. 2022; Olden and Poff 2004). In contrast, functional studies track richness and convergence/divergence in the trait space that shape assembly via niche mechanisms such as environmental filtering and competitive exclusion (Cantera et al. 2023; Suárez-Tangil and Rodríguez 2023). Together, these complementary indicators provide a more holistic view of human impacts on communities.

Broad-scale analyses also often overlook the critical role of geographic heterogeneity in shaping wildlife responses to human pressure, even though the type and intensity of these pressures differ significantly across biomes. In temperate and subalpine regions, long-standing human activities like forest grazing have altered wildlife composition in forests for centuries (Borman 2005; Varga et al. 2020). The long history of human activity in these regions likely already filtered out the most sensitive species, so contemporary surveys capture mainly tolerant assemblages and understate true impacts (Newbold et al. 2020). In contrast, low-latitude tropical communities tend to be more sensitive to environmental change and thus more vulnerable to modern human pressures (Betts et al. 2019; Turvey and Fritz 2011). At the same time, tropical forests are being lost at alarming rates, placing their wildlife at grave risk (Greco et al. 2025; Myers 1983). In addition to current stressors, historical human pressures may produce delayed effects on biodiversity patterns (Hylander and Ehrlén 2013). Given that past urbanization and land-use dynamics are closely associated with contemporary species distributions (Dullinger et al. 2013; Polaina et al. 2019; Yang et al. 2021), their effects on wildlife should be quantified using more temporally explicit covariates like the rate of change in building-density.

To date, limitations in data quality and geographic sampling coverage have hindered rigorous, objective assessments of forest wildlife status. Large-scale evaluations of mammalian composition and functional diversity often rely on range maps, which can confound coarse distributions with local habitat use and introduce biases (Arbogast and Kerhoulas 2024; Baisero et al. 2020; Cooke, Bates, and Eigenbrod 2019; Hedberg and Smith 2025). In contrast, camera trapping provides site-level occurrence data that are better suited for quantifying human impacts on species composition and functional trait space in mammal communities and are widely used to test biodiversity hypotheses (Boron et al. 2019; Coronel-Arellano et al. 2018; Greco et al. 2025; Lewis et al. 2021).

Notably, several foundational studies have laid the groundwork for understanding mammal diversity patterns through integrated taxonomic, functional, and phylogenetic approaches. Penone et al. (2016) conducted a global analysis of mammal beta diversity, revealing how taxonomic, phylogenetic, and trait dimensions of diversity respond to historical isolation and environmental similarity, highlighting the value of multi-dimensional diversity assessments for disentangling assembly mechanisms. Oliveira et al. (2016) further demonstrated that species richness and functional diversity of mammals accumulate through distinct processes—with species richness shaped by equilibrium environmental factors and functional diversity driven by non-equilibrium evolutionary time—underscoring the need to jointly consider these diversity facets when evaluating global change impacts. Meanwhile, Feng et al. (2023) focused on Chinese forest mammals, identifying elevation range and contemporary climate as key drivers of taxonomic, functional, and phylogenetic diversity, and emphasizing the importance of integrating natural environmental gradients into diversity response analyses. Despite these advances, tropical Asia remains underrepresented in such broad-scale integrated studies, with most research concentrated on temperate or Neotropical regions. Our study builds on these prior works by focusing on Southeast Asia—a biodiversity hotspot facing intense anthropogenic pressure—and integrating compositional homogenization and functional trait responses to human disturbances across biomes, thereby filling a critical geographic gap in understanding region-specific mammal community dynamics under global change.

Southeast Asia's biomes exhibit striking heterogeneity in elevation (from sea level to >5000 m in the Hengduan Mountains), terrain complexity (from flat lowland floodplains to rugged montane landscapes), and climate (from seasonal temperate zones with freezing winters to aseasonal tropical zones with year-round high rainfall or pronounced dry seasons). These gradients directly shape species pools and functional trait distributions: Temperate Coniferous Forests (TCF) are characterized by coniferous vegetation, low temperatures, and high topographic complexity, supporting cold-tolerant species with broader climatic niches; Tropical & Subtropical Moist Broadleaf Forests (TSMBF) feature high rainfall, dense evergreen vegetation, and high endemism, with species adapted to stable but narrow ecological conditions. As highlighted by Rovero et al. (2020), such biome-specific habitat structure and species adaptations translate to varying vulnerability to human disturbance: tropical biomes tend to exhibit higher functional vulnerability due to narrower climatic niches and greater dependence on intact

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habitat structure, while temperate biomes may be buffered by more resilient species pools and complex topography that creates refugia. Penone et al. (2016) further noted that biome-specific environmental filters drive beta diversity patterns, with tropical forests across different realms showing convergent functional traits despite taxonomic distinctness. This pattern underscores the importance of testing how local human pressures interact with biome-specific filters to reshape communities. Mazel et al. (2017) observed that ecological niche evolution in mammals is strongly tied to biome characteristics, with assemblages in similar biomes (even across regions) evolving convergent functional strategies, while assemblages within the same biome but under different disturbance regimes may diverge sharply. By sampling across these two biomes within a single biogeographic region, we avoid confounding biogeographic history with biome effects, enabling clear tests of how human pressures interact with natural environmental gradients to shape compositional and functional patterns.

Here, we address three questions: (1) How do mammal community compositional and functional diversity patterns shift in response to different human pressures? (2) How do compositional and functional diversity patterns relate to one another, and how do they jointly inform community assembly processes? (3) To what extent do geographic and climatic conditions mediate mammal community responses to human pressures? We compiled camera-trap data from 4508 forest sites across Southeast Asia, spanning a broad latitudinal gradient and predominantly distributed within two biomes: TCF and TSMBF. Both of these forest ecoregions are likely to be affected by climate change or human activities in the future, particularly the TSMBF with its rich species diversity (Olson and Dinerstein 1998), where the probability of impact will increase regardless of whether it is in a critical or intact conservation state (Wang et al. 2019). We justified our geographic design (sampling across these biomes) by noting that it allows for testing how human pressure effects are mediated by natural environmental context—a key knowledge gap for tropical Asian mammals. Relative to TCF, study areas in TSMBF are predominantly located in low-elevation, gently sloping areas, characterized by higher precipitation and greater net primary productivity. We considered multiple landscape change indicators: habitat fragmentation (patch density, PD), human modification (HM), and historical building-density change (BC), together with spatial proxies of human–wildlife interactions: human presence (relative abundance index for humans, RAIH) and grazing (relative abundance index for livestock, RAIL). Using the median values of these indicators as benchmarks, we delineated impacted communities from reference states. Following Keck et al. (2025), we computed the log-response ratio of homogeneity (LRR homogeneity) and the log-response ratio of shift (LRR shift) to evaluate trends in compositional homogenization/heterogenization within impacted communities relative to natural scenarios. We also quantified functional richness (FRic) and its standardized effect size (SES.FRic), functional dispersion (FDis), and functional evenness (FEve) to assess convergence and divergence trends in functional assembly. We tested four hypotheses about how human pressures influence mammal community assembly.

i. *H1*. Human pressures favor species that tolerate or exploit disturbance and, at the same time, promote ecological niche differentiation among surviving or newly established taxa (Sévêque et al. 2020). As a result, species composition among impacted communities is expected to become more similar (compositional homogenization), while the functional structure of communities is expected to become more divergent. We expect this pattern to be strongest in TSMBF, where high resource availability may amplify competitive exclusion and niche partitioning. In contrast, TCF may show weaker responses due to lower species density and more heterogeneous terrain that mitigates homogenization (Figure 1a, H1).

ii. *H2*. Human pressures may reduce the viability of some species and create competitive imbalances among ecologically similar taxa, producing range contractions or local extinctions for affected lineages (Lesmeister et al. 2015; Michalski and Peres 2005). Because environmental contexts and population sizes differ among study areas, these losses can occur stochastically across sites (Smith and Ameida 2020), leading to increased compositional heterogeneity and functional divergence. This pattern is predicted to be most pronounced in TCF, where complex topography and stochastic post-disturbance survival amplify among-site differences, whereas TSMBF (with more ecologically similar species pools) may show less stochasticity in species loss (Figure 1b, H2).

iii. *H3*. If human pressures act as a strong environmental filter—removing species that cannot tolerate disturbance while selectively retaining species with advantageous traits (Kraft et al. 2015)—then impacted communities should become more compositionally homogeneous and exhibit functional convergence (Figure 1c). We anticipate this pattern to be strongest in TSMBF, where intense disturbance (e.g., high building-density, grazing) may select for a narrow set of disturbance-tolerant traits (e.g., small body size, omnivory), whereas TCF may show weaker convergence due to a more diverse set of refugia and trait strategies (Figure 1c, H3).

iv. *H4*. Because geographic, climatic, and historical disturbance regimes differ among biomes, the effects of human pressures are expected to vary across biomes, leading to biome-specific responses in community composition and functional structure. We predicted that mammal communities in TSMBF would be consistent with H3, owing to the biome's relatively stable climatic regime and homogeneous natural selection pressures, which had given rise to similar mammalian assemblage structures (Jetz and Fine 2012; Mazel et al. 2017; Penone et al. 2016). Human pressures such as uncontrolled hunting and habitat loss-driven species-specific extinctions may trigger functional responses analogous to convergent evolution (Rovero et al. 2020). Additionally, mammal communities in TCF were expected to support H2. This biome, distributed across low-latitude mountains in Southeast Asia, features extensive elevational ranges and heterogeneous habitats which are primary contributors to species diversity (Feng et al. 2023). Anthropogenic disturbances can amplify spatial isolation among local communities, thereby promoting ecological divergence, intensifying interspecific competition and driving diversification (Ålund et al. 2023; Reyne et al. 2023).

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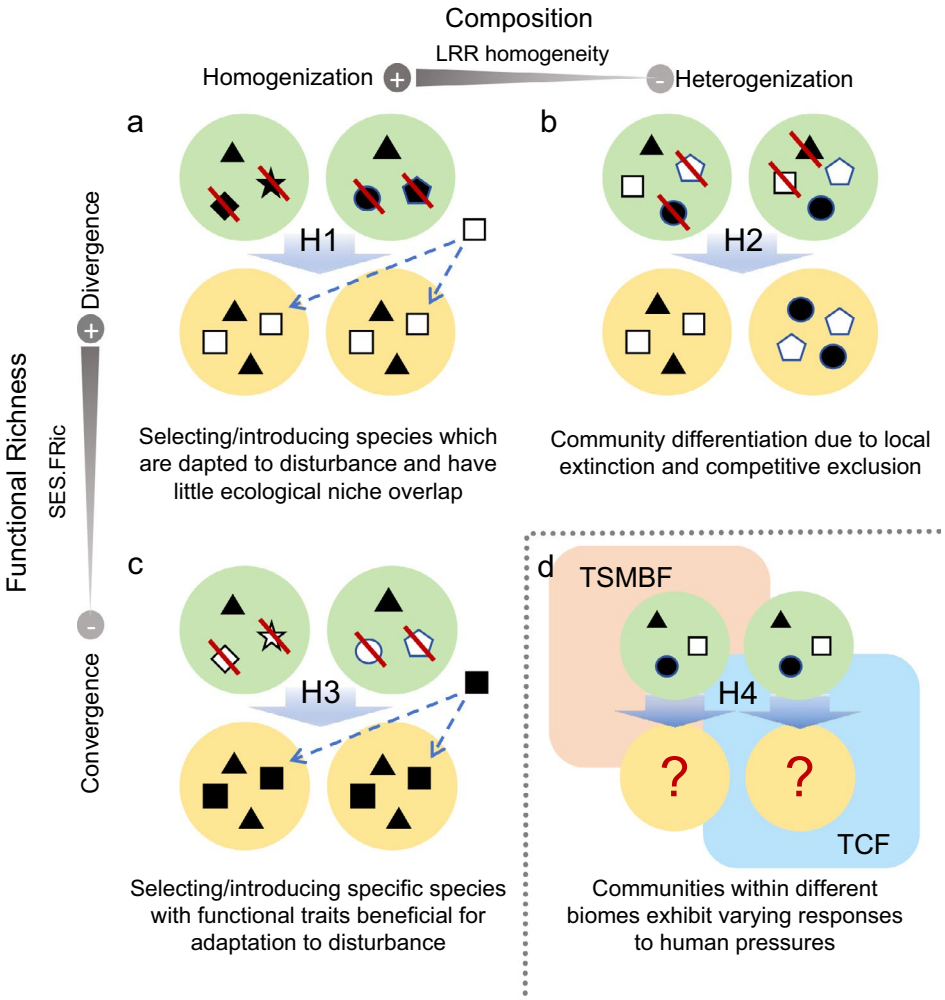


FIGURE 1 | Hypothesized processes (H1–H4) by which human pressures influence community composition and functional patterns. Compositional homogenization and functional convergence are evaluated by log-ratio response homogeneity (LRR homogeneity) and the standard effect size of functional richness (SES.FRic), respectively. The figure depicts four predicted scenarios: Compositional homogenization with divergent functional patterns (a, H1), compositional heterogenization with divergent functional patterns (b, H2), compositional homogenization with convergent functional patterns (c, H3), and various responses to human pressures from communities within different biomes (d, H4). The green and yellow backgrounds represent reference and impacted communities, respectively. Different shapes denote distinct species, whereas different colors represent different types of functional traits. Red lines indicate the disappearance of a species from the community, and dashed lines represent the introduction or spread of species into the community.

Our study connects compositional and functional responses to different types of anthropogenic disturbance and reveals how these effects vary synergistically across biomes, providing new insights into the large-scale impacts of humans on ecosystems and biodiversity. This empirical investigation, based on broad-scale camera trapping data, illustrates how combining compositional and functional indicators yields a more comprehensive understanding of anthropogenic impacts on biodiversity.

2 | Methods

2.1 | Study Area and Camera Trap Dataset

We compiled single-season (November–March) camera trap data from forest ecosystems in Southeast Asia collected between 2010 and 2024, spanning 8°27′S–29°38′N and 82°53′E–117°46′E (Figure 2; Table S1). The data sources included our own surveys ($N = 43$), CamTrapAsia ($N = 18$) (Mendes et al. 2024), Wildlife Insights ($N = 11$) (Ahumada et al. 2020), and eMammal ($N = 2$) (McSheaet al. 2016). Study areas were placed ≥ 10 km apart unless separated by geographic barriers (e.g., rivers, mountains). The minimum distance between study area was 15.76 km. To adequately capture the local species pool across tropical regions, we implemented a minimum of 30 camera trap sites within each study area of the selected projects (Kays et al. 2020). The minimum distance between camera trap sites was 1045 m (range = 1000–2000 m).

In total, we collated species distribution observations from 74 study areas and 4508 camera trap sites (mean = 60.92 per area; range = 30–290), totaling 360,328 camera trap days (mean = 4869 per area; range = 2016–26,680). Each area recorded ≥ 5 mammal species (mean = 16.09 per area). Sites span nine countries and broad natural–anthropogenic gradients within three biomes: TSMBF ($N = 40$), TCF ($N = 28$), and tropical & subtropical dry

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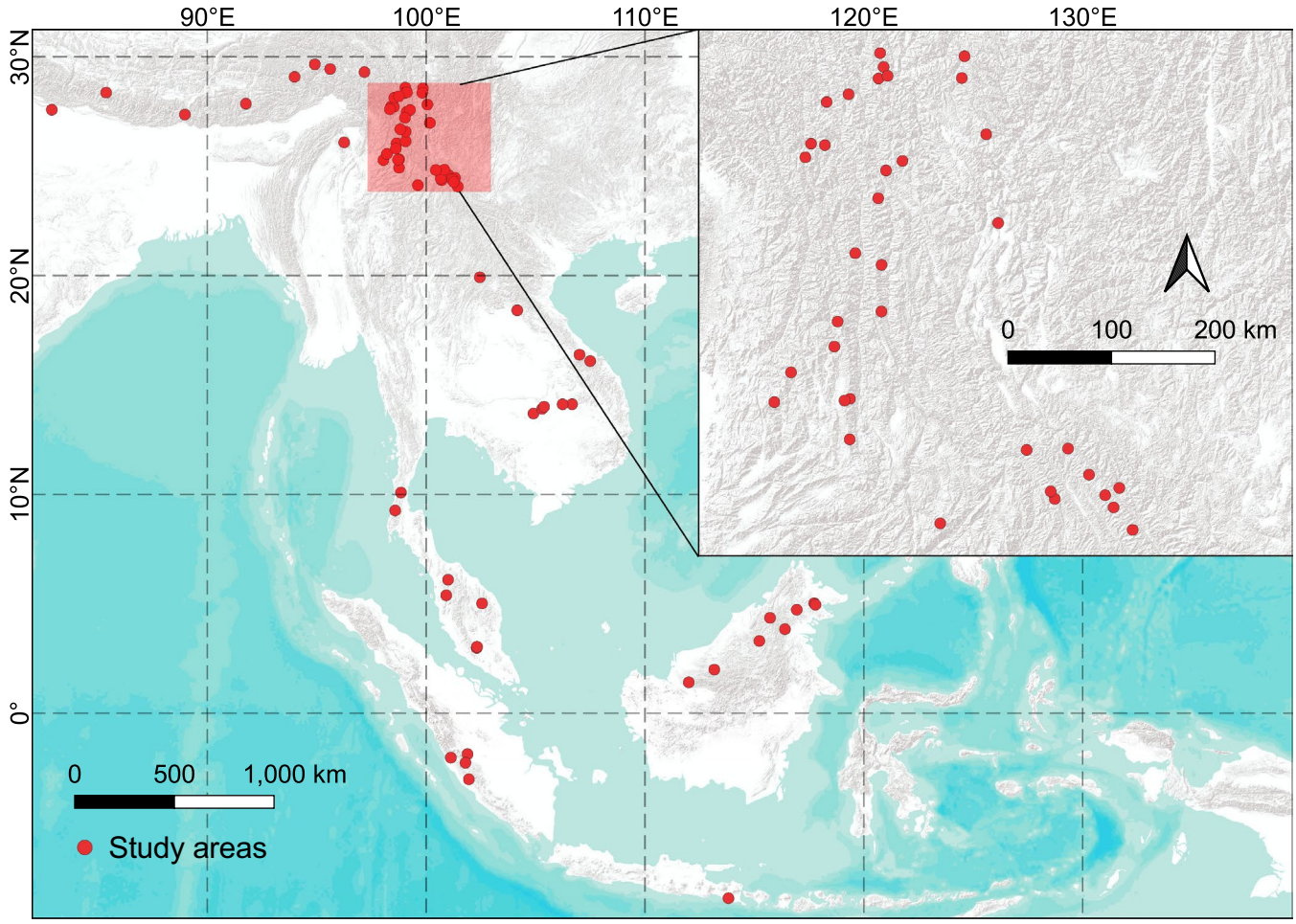


FIGURE 2 | Locations of the study areas in Southeast Asian forests. A map of the 74 study areas included in this study. The inset shows the study areas located in the Hengduan Mountains of China.

broadleaf forests (TSDBF, $N = 6$). To disentangle human pressure effects from natural biome heterogeneity in a single biogeographic region, we conducted separate analyses for mammal communities in TSMBF and TCF, alongside a pooled analysis of the entire dataset. These two biomes are highly representative of tropical and subtropical regions in Southeast Asia. The number of study areas in TSDBF was insufficient to support robust comparative analyses.

2.2 | Covariates: Description and Processing

To capture human pressures and natural attributes, we derived 11 covariates. For covariates requiring spatial aggregation, we evaluated buffers of 1, 5, and 10 km and assessed their pairwise Pearson correlations using $|r| = 0.7$ as the collinearity threshold. Strong correlations were evident between 1 and 5 km and between 5 and 10 km (Figure S4a); therefore, we adopted a 5-km radius for subsequent analyses, which appropriately reflects local habitat change and human activity affecting medium- and large-bodied mammals (Haskell et al. 2002; Moll et al. 2020). We further ensured that multicollinearity among continuous covariates remained below $|r| = 0.7$ (Figure S4b). For each study area, we calculated camera-trap-array means for all the environmental covariates and standardized them (mean = 0, SD = 1). Where camera trap array coordinates were unavailable, we randomly generated 1000 points within the study area to compute mean values and standardized those covariates accordingly.

2.3 | Human Activity Covariates

We quantified four dimensions of human pressure. (i) Habitat fragmentation was indexed by patch density (PD) computed with landscapemetrics v2.1.4 (Hesselbarth et al. 2019) from ESA WorldCover 10 m 2021 v200 land cover (Zanaga et al. 2022) after masking to camera trap array extent. PD was then calculated on a site-centered buffer and summarized per site. (ii) Human modification (HM) represented the composite extent of human alteration using the Global Human Modification of Terrestrial Systems v1 (2016; 1 km) surface (Kennedy et al. 2020). (iii) To capture historical pressures, we derived building-density change (BC) over > 30 years for each study area as the difference between building-density during the camera-trap deployment period and building-density circa 1980, using the GHS-BUILT-S (100 m) grids (Potapov et al. 2022). (iv) We quantified direct human presence with the relative abundance index for humans (RAIH) and grazing pressure with the relative abundance index for livestock (RAIL), computed as detections per 100 camera-trap nights (RAI = 100 × detections/trap-nights).

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2.4 | Natural-Environment Covariates

To account for the natural environment context, we assembled topographic, climatic, productivity, vegetation change, and biome covariates. (i) Altitude (Alt) and terrain roughness (Rough), key contributors to vertebrate community assembly (García-Llamas et al. 2019), were extracted from NASADEM_HGT v001 (1 arc-s) (NASA JPL 2020). (ii) Precipitation of the driest month (Bio14), relevant to forest microclimate and stability (Roy 2011; Xu et al. 2022), came from WorldClim v2 bioclimatic variables (1 arc-s) (Fick and Hijmans 2017). (iii) Net primary productivity (NPP), a putative driver of richness and biodiversity patterns (Willig et al. 2003), was taken from MOD17A3HGF v061 (500 m) (Running and Zhao 2021) and averaged across years overlapping the sampling period. (iv) We characterized long-term vegetation dynamics with the rate of tree-cover change (TC) from Long-term Global Land Change (1982–2016; 0.05 degree) (Song et al. 2018). (v) Finally, to contextualize regional biological communities and partially capture broad-scale biodiversity patterns (Smith et al. 2018), we assigned each site to a biome following Ecoregions 2017 (Dinerstein et al. 2017), focusing on Tropical & Subtropical Moist Broadleaf Forests (TSMBF) and Temperate Coniferous Forests (TCF), where sample sizes were sufficient, for comparative analyses.

2.5 | Functional Traits

We assembled a trait database for all the monitored species. Body mass, diet, circadian rhythm, and foraging stratum data were obtained from EltonTraits (Wilman et al. 2014); litter size data were obtained from COMBINE (Soria et al. 2021); and habitat breadth data were obtained from the IUCN Red List (IUCN 2024). These traits represent spatiotemporal resource use and are commonly applied to characterize mammalian functional diversity (Cooke, Eigenbrod, and Bates 2019).

To align with camera trap detectability, we excluded wild mammal species with a mean body mass < 500 g and strictly arboreal species (Anile and Devillard 2016; Rich et al. 2016). The target taxa thus comprised ground-dwelling or semiarboreal, medium- to large-bodied mammals. Taxonomy followed GBIF (Scott et al. 2024). For taxa not identifiable to species from imagery (e.g., *Herpestes* spp., *Melogale* spp., *Muntiacus* spp., *Mustela* spp., *Rhizomys* spp., and *Tragulus* spp.), we retained genus -level assignments. The final dataset included 317,710 occurrences of 112 mammalian categories spanning 10 orders, 26 families, and 68 genera.

The trait transformations followed standard practice (Villéger et al. 2008): log-transformed body mass and litter size; square root habitat breadth; and standardizing all continuous traits (mean = 0, SD = 1). For the genus-level records, we averaged body mass, diet, habitat breadth, and litter size over co-occurring congeneric species within the sampled areas (based on IUCN ranges) (IUCN 2024) and adopted the common state for circadian rhythm, foraging stratum, and protection level.

2.6 | LRR Homogeneity and Compositional Shift

We quantified how community composition changes under human pressures using two log-response–ratio (LRR) indices adapted for multivariate community data: LRR homogeneity and LRR shift (Keck et al. 2025). For each study area, we pooled species occurrence data across all camera trap sites and treated them as a community. Environmental covariates for each study area were represented by the average values of environmental factors calculated across all camera sites within that area. We first split the study areas into impacted and reference groups using the median of human pressure covariate across these study areas as the threshold and then computed pairwise Euclidean distances of community composition. We calculated LRR of impacted to reference communities for compositional similarities (homogeneity: LRR homogeneity) (Equation 1). Next, we examined the change in species composition between impacted and reference communities quantified as compositional shift (LRR shift) (Equation 2).

$$\text{LRR homogeneity} = -\ln\left(\overline{D_I} / \overline{D_R}\right) \tag{1}$$

$$\text{LRR shift} = \ln\left(\overline{D_B} / \overline{D_W}\right) \tag{2}$$

$\overline{D_I}$ and $\overline{D_R}$ are the average distances in impacted and reference communities, respectively. Multiplying the LRR of $\overline{D_I}$ and $\overline{D_R}$ by −1 expresses the change in community similarity more clearly. LRR homogeneity values > 0 indicates higher similarity of impacted communities compared to reference ones, and vice versa. $\overline{D_B}$ is the average distance between impacted and reference communities while $\overline{D_W}$ is the average distance in the impacted and reference communities. LRR shift values > 0 indicate greater inter-group differences compared with intra-group differences, and vice versa. Because environmental structure and species–occurrence constraints can influence community distances, we standardized both LRR indices with null models to obtain standardized effect sizes (SES) and confidence intervals. Communities were grouped by biome, and the community matrix was randomized 999 times using the independent-swap algorithm (maintaining site richness and species frequencies) implemented in the R package picante (Kembel et al. 2010); this produced 90% and 95% CIs for both LRR homogeneity and LRR shift. We considered 95% CIs excluding zero as statistically significant and 90% CIs excluding zero as informative evidence. As the null distributions were approximately normal (see Figures S5, S6), we used the null model means in SES calculations.

2.7 | Functional Diversity

We computed pairwise Gower distances among species using gower (v1.0.1) (van der Loo 2022) with equal trait weights and then ordinated the species with PCoA using ape (v5.8) (Paradis et al. 2004). To analyze functional diversity across communities, we constructed 5-dimensional trait spaces (explaining 80.36% of total variation) using the hypervolume methods (Mammola and Cardoso 2020; Soares et al. 2022) implemented in BAT (v2.9.6) (Cardoso et al. 2015). We then calculated functional richness (FRic), functional dispersion (FDis), and functional evenness (FEve) (Carmona et al. 2016, 2019; Laliberté and Legendre 2010; Mammola and Cardoso 2020). FRic represents the niche space volume occupied by community members (Legras et al. 2018); FDis quantifies trait dissimilarity among species (Cooke, Bates, and Eigenbrod 2019); and FEve measures the regularity of

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species' trait distributions (Ricotta et al. 2014). Because camera trap effort varied across the study areas, we based the functional metrics on occurrence (presence) only, not detection frequency. To infer convergence/divergence in functional assembly, we randomized community composition ($N = 999$) to generate null expectations and derived SES.FRic. SES.FRic < 0 indicates convergence; SES.FRic > 0 indicates divergence.

2.8 | Statistical Modeling

We fitted generalized linear models (GLMs) with functional diversity indices as responses and environmental covariates as predictors, including the number of camera trap sites per study area as an offset to account for the unequal sampling effort among sites. When residual diagnostics suggested curvature (parabolic trends), we added quadratic terms for the corresponding predictors. Model support was assessed using the Akaike information criterion (AIC) (Bozdogan 1987), with stepwise regression to identify the minimum-AIC model. We plotted the predicted curves of SES.FRic for important human pressure covariates with the mean values of non-target covariates. All analyses were conducted in R v4.4.2 (Team 2014).

3 | Results

3.1 | Human Impacts on Compositional Homogeneity and Shifts

Across all the study sites, compositional homogeneity increased significantly under intensive human modification (LRR homogeneity = 7.940, 95% CI = 5.891–9.896), human presence (LRR homogeneity = 6.01, 95% CI = 4.662–8.605), and building-density change (LRR homogeneity = 2.802, 95% CI = 0.849–4.837) (Figure 3a). There was no difference in homogeneity across patch density classes and grazing classes. When analyses were restricted to TSMBF, the homogeneity increased significantly in the study areas with high-frequency grazing (LRR homogeneity = 5.188, 95% CI = 3.027–6.945), and the effects of other human pressures were consistent across sites. In contrast, within TCF, heterogenization occurred under intense human presence (LRR homogeneity = −3.925, 95% CI = −5.889–−1.980), human modification (LRR homogeneity = −1.821, 95% CI = −3.854–−0.106), habitat fragmentation (LRR homogeneity = −2.834, 95% CI = −4.831–−0.812), and great changes in building-density (LRR homogeneity = −4.261, 95% CI = −6.440–−2.321).

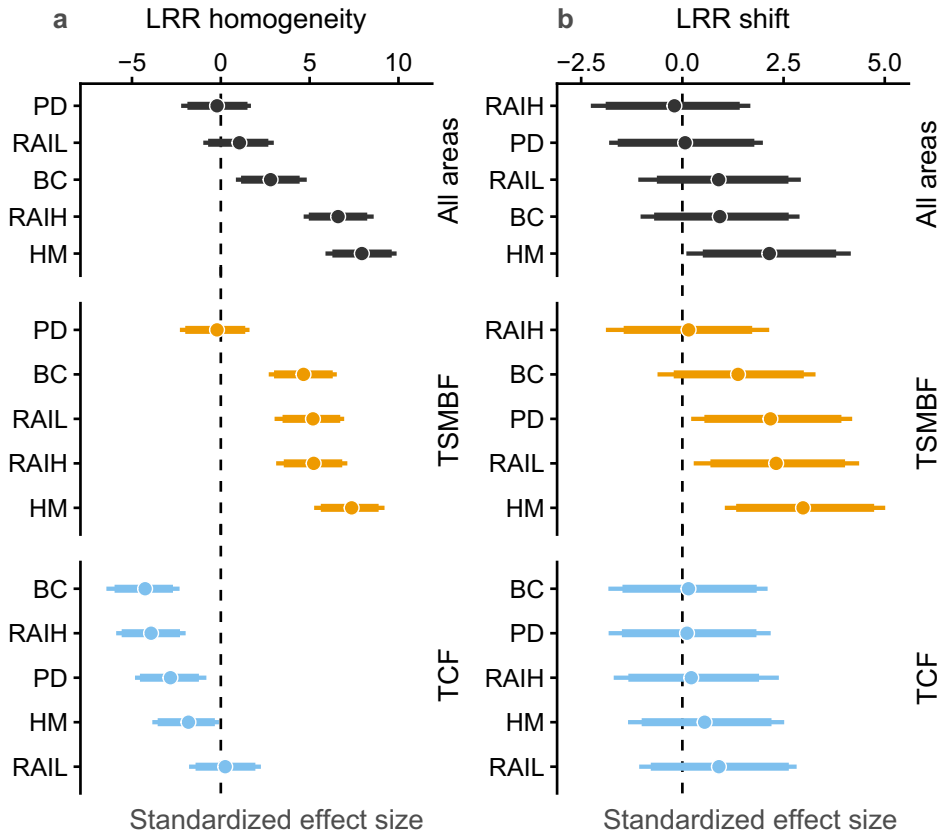


FIGURE 3 | Impacts of human pressures on the homogeneity and shifts of mammal community composition. (a) Standardized effect sizes of LRR homogeneity, where values greater than zero indicate compositional homogenization and values less than zero indicate compositional heterogenization. (b) The standardized effect sizes of LRR shift, with values greater than zero indicating differences in species composition between the impacted and reference communities. PD, patch density; HM, human modification; BC, building-density change; RAIH, relative abundance index for humans; RAIL, relative abundance index for livestock. The results are organized by biomes, including all study areas (all areas), Tropical & Subtropical Moist Broadleaf Forests (TSMBF), and Temperate Coniferous Forests (TCF). The dots represent the standardized effect sizes computed from the null model, with larger bars denoting 90% confidence intervals and thinner bars representing 95% confidence intervals.

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The LRR shift indicated marked compositional differences between highly human-modified communities versus the reference communities (LRR shift = 2.148, 95% CI = 0.099–4.161), whereas there was no variation across other human pressure gradients. In TSMBF, significant shifts were also observed for habitat fragmentation (LRR shift = 2.176, 95% CI = 0.217–4.195) and grazing (LRR shift = 2.317, 95% CI = 0.280–4.367), in addition to human modification (LRR shift = 2.980, 95% CI = 1.049–5.011). No compositional shifts were detected in the TCF for any human pressure variable.

3.2 | Human Impacts on Functional Diversity

The effects of environmental covariates on FRic, FDis, and FEve are shown in Figure 4 and Table S3. Habitat fragmentation had a pronounced positive effect on FRic across all study areas (estimate = 0.521, 95% CI = 0.073–0.972, 2.94% variance explained) and within TCF (estimate = 1.107, 95% CI = 0.122–1.960, 7.32% variance explained). In contrast, FRic and human presence exhibited a negative correlation (all study areas: estimate = −0.297, 95% CI = −0.546–−0.043, 3.22% variance explained; TCF: estimate = −0.426, 95% CI = −0.771–−0.066, 9.24% variance explained). In the TSMBF, FRic showed a slight positive correlation with human modification (estimate = 0.215, 90% CI = 0.002–0.416), explaining 5.18% of the variance.

FDis was similarly positively associated with habitat fragmentation (estimate = 0.046, 95% CI = 0.030–0.062), accounting for 28.26% of the explained variance. In TCF, FDis also demonstrated a positive correlation with habitat fragmentation (estimate = 0.029, 90% CI = 0.004–0.053, 4.44% variance explained) and declined with increasing human modification (estimate = 0.012, 95% CI = −0.052–−0.004, 7.74% variance explained). By contrast, FDis tended to be negatively correlated with habitat fragmentation (estimate = −0.013, 90% CI = −0.026–0.001, 6.04% variance explained) and positively correlated with human modification (estimate = 0.012, 90% CI = 0.004–0.020, 20.38% variance explained) in TSMBF.

FEve showed a positive but nonsignificant association with habitat fragmentation across the study areas (estimate = 0.008, 90% CI = 0.001–0.016), accounting for 7.13% of the variance. In TSMBF, FEve was positively associated with human modification (estimate = 0.012, 95% CI = 0.004–0.019; 27.88% variance) and grazing (estimate = 0.011, 95% CI = 0.002–0.021; 27.57% variance). For TCF, FEve was positively correlated with building-density change (estimate = 0.013, 95% CI = 0.005–0.022;

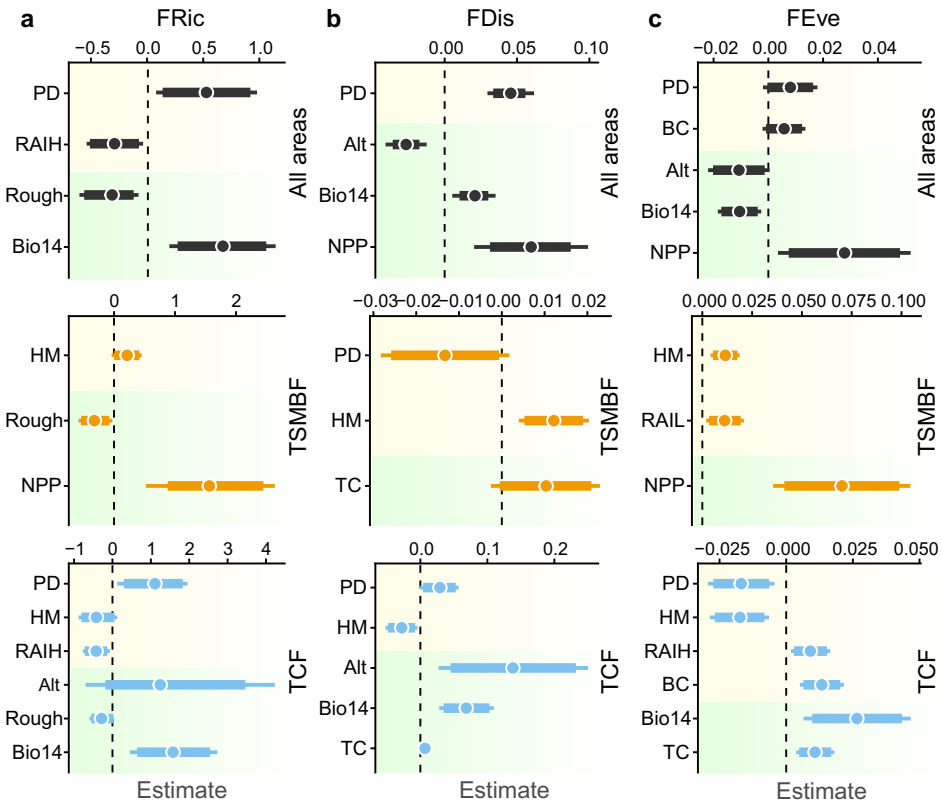


FIGURE 4 | Effects of environmental covariates on the functional diversity of mammal communities. These plots show the associations between environmental covariates and functional diversity: Functional richness (a), functional dispersion (b), and functional evenness (c). The results are organized by biomes, including all study areas (All areas), Tropical & Subtropical Moist Broadleaf Forests (TSMBF), and Temperate Coniferous Forests (TCF). PD, patch density; HM, human modification; BC, building-density change; RAIH, relative abundance index for humans; RAIL, relative abundance index for livestock; Alt, altitude; Rough, terrain roughness; Bio14, precipitation of driest month; NPP, net primary productivity; TC, tree-cover change. The dots represent the coefficient estimates of the linear terms of the covariates, with larger bars representing 90% confidence intervals and thinner bars denoting 95% confidence intervals. The yellow backgrounds denote human pressures, and the green backgrounds denote the natural attributes of the study areas.

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14.86% variance) and human presence (estimate = 0.009, 95% CI = 0.002–0.016; 11.64% variance) and negatively correlated with habitat fragmentation (estimate = −0.017, 95% CI = −0.029–−0.004; 15.61% variance) and human modification (estimate = −0.017, 95% CI = −0.029–−0.006; 7.24% variance).

Human pressures were strongly associated with SES.FRic (Figure 5; Table S3). Across all study areas, human modification, building-density change, and habitat fragmentation were the top contributors to SES.FRic variation, explaining 30.06%, 26.48%, and 19.09% of the variance, respectively. The functional pattern was divergent with human modification (estimate = 0.351, 95% CI = 0.137–0.559) and habitat fragmentation (estimate = 0.425, 95% CI = 0.187–0.626). In contrast, the association with building-density change was nonlinear, diverging at medium values but converging at extremes (estimate = −0.283, 95% CI = −0.420–−0.128). In TSMBF, SES.FRic also increased with human modification (estimate = 0.404, 95% CI = 0.168–0.633; explained variance = 21.14%) and habitat fragmentation (estimate = 0.318, 95% CI = 0.0223–0.589; explained variance = 8.89%), indicating a transition from convergence to divergence. On the contrary, the pattern shifted from divergence to convergence with increasing building-density change (estimate = −0.329, 95% CI = −0.620–−0.031; explained variance = 4.08%). In TCF, the functional pattern's divergence weakened as grazing intensity increased (estimate = −1.240, 90% CI = −2.299–−0.154; explained variance = 7.39%). Additionally, it exhibited a hump-shaped relationship with building-density change (estimate = −1.002, 95% CI = −1.800–−0.265; explained variance = 6.610%). and habitat fragmentation (estimate = −1.033, 95% CI = −2.189–−0.026; explained variance = 4.67%).

4 | Discussion

We have empirically demonstrated that human pressures profoundly shape the assembly of forest-mammal communities in Southeast Asia from local to regional scales, and we provide a comprehensive assessment of complementary biodiversity indicators—an approach rarely taken previously. All the human pressures examined altered compositional and functional patterns, albeit through different processes. Overall, human pressures tended to homogenize community composition at large spatial scales, but the strength and direction of effects varied by

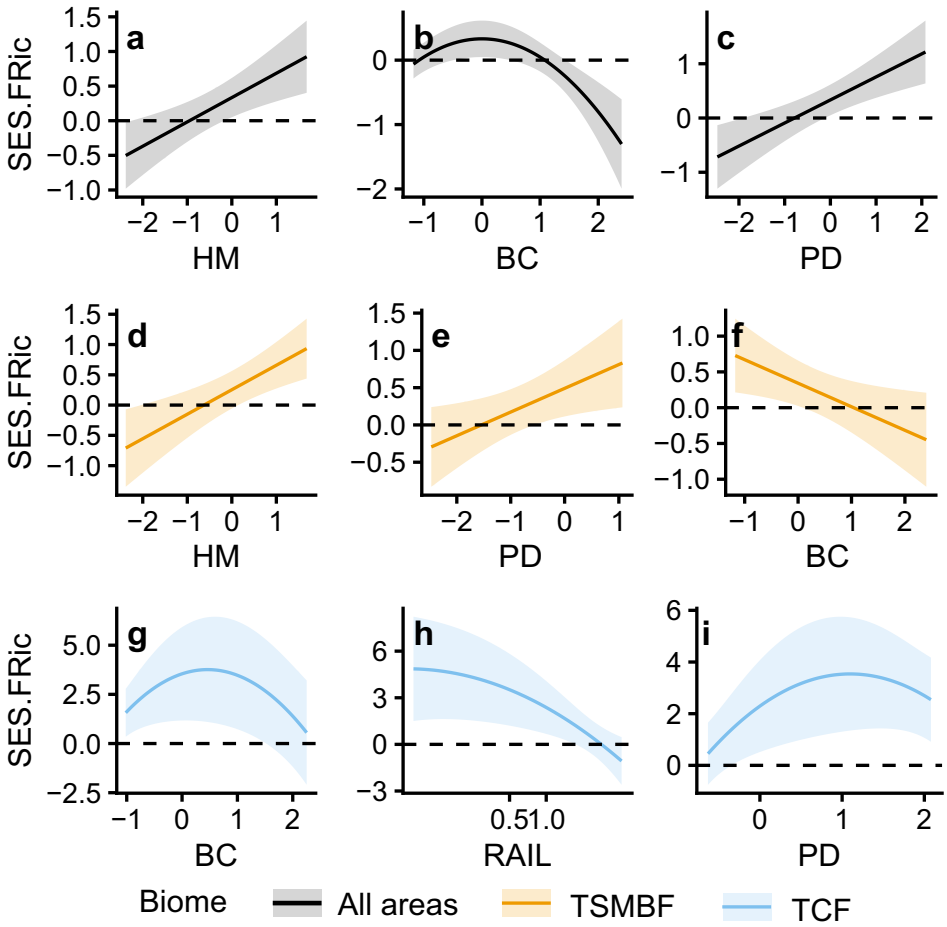


FIGURE 5 | Impacts of human pressures on the functional patterns of mammal communities. These predicted curves illustrate the standardized effect sizes of functional richness (SES.FRic) in response to human pressures. The results are organized by biomes, including all study areas (All areas, a–c), Tropical & Subtropical Moist Broadleaf Forests (TSMBF, d–f), and Temperate Coniferous Forests (TCF, g–i). PD, patch density; HM, human modification; BC, building-density change; RAIH, relative abundance index for humans; RAIL, relative abundance index for livestock. The shaded areas represent 95% confidence intervals. A SES.FRic value greater than zero indicates functional divergence; otherwise, it signifies functional convergence.

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biome. As strong predictors of SES.FRic, human pressures also contribute to functional community structure, with comparable importance to that of long-standing natural covariates that influence wildlife evolution.

Our results align with and extend key findings from Oliveira et al. (2016) and Feng et al. (2023), bridging global patterns of mammal diversity with regional dynamics in Southeast Asia. Oliveira et al. (2016) highlighted that mammal species richness and functional diversity are decoupled globally, with species richness driven by environmental factors (e.g., productivity) and functional diversity shaped by evolutionary time and niche conservatism. Our study confirms this decoupling in the context of anthropogenic disturbance: while human pressures (e.g., habitat fragmentation, human modification) drive compositional homogenization (a taxonomic pattern), functional diversity responds independently—showing either divergence or convergence depending on disturbance type and biome. For instance, high-intensity human modification promotes functional divergence in Southeast Asian mammals, which extends Oliveira et al.'s (2016) observation of functional redundancy in species-rich regions by demonstrating how anthropogenic stressors can disrupt this redundancy and drive trait differentiation among surviving species.

Feng et al. (2023) identified elevation range and contemporary climate as key drivers of mammal taxonomic, phylogenetic, and functional diversity in Chinese forests, supporting the habitat heterogeneity and energy availability hypotheses. Our findings reinforce these global drivers while highlighting their interaction with human pressures in Southeast Asia. Specifically, we show that habitat heterogeneity (indexed by patch density) enhances functional richness and dispersion across our study areas, consistent with Feng et al.'s (2023) habitat heterogeneity hypothesis. However, we further demonstrate that human pressures can modulate these natural drivers: in Tropical & Subtropical Moist Broadleaf Forests (TSMBF), fragmentation amplifies functional divergence beyond the baseline effects of natural heterogeneity, whereas in Temperate Coniferous Forests (TCF), it interacts with topographic complexity to produce compositional heterogenization. This regional nuance expands Feng et al.'s (2023) insights by illustrating how anthropogenic disturbance can reshape diversity patterns even in systems structured by strong natural gradients. Notably, our results for tropical versus temperate biomes echo Oliveira et al.'s (2016) observation of biome-specific functional diversity patterns. TSMBF in Southeast Asia—consistent with the Neotropics in Oliveira et al.'s (2016) study—exhibits high species richness but heightened vulnerability to anthropogenic functional shifts, likely due to niche conservatism and narrow climatic tolerances of tropical species. In contrast, TCF shows more resilient functional patterns, mirroring the role of evolutionary time in shaping functional diversity (Oliveira et al. 2016) and the buffering effect of topographic complexity (Feng et al. 2023). Together, these connections underscore that Southeast Asian large mammals follow global diversity dynamics but respond uniquely to regional anthropogenic pressures, highlighting the need for context-specific conservation strategies.

For Tropical & Subtropical Moist Broadleaf Forests (TSMBF), we observed the strongest compositional homogenization and functional shifts—consistent with our prediction of heightened vulnerability in tropical biomes. This finding mirrors Rovero et al.'s (2020) global assessment of tropical forest mammals, which highlighted that tropical communities exhibit higher functional vulnerability due to narrower climatic niches, greater endemism, and dependence on intact habitat structure. Penone et al. (2016) further noted that tropical forests across different realms show low trait beta diversity despite high phylogenetic beta diversity, a pattern driven by convergent functional strategies in response to similar environmental filters (e.g., stable warm climates, high resource availability). In our Southeast Asian TSMBF, this translates to a high sensitivity to human disturbance: intensive human modification and building-density change drive the replacement of endemic specialists (e.g., banded palm civet, sun bear) with widespread generalists (e.g., wild boar, masked palm civet), resulting in pronounced compositional homogenization and functional convergence at high disturbance levels. This aligns with Mazel et al.'s (2017) insight that assemblages in similar biomes (even across regions) evolve convergent functional traits, and that anthropogenic pressures can amplify this convergence by filtering out niche-specialized species.

For Temperate Coniferous Forests (TCF), we detected more variable responses to human pressures, including compositional heterogenization under fragmentation and building-density change, supporting our hypothesis that temperate biomes are buffered by more resilient species pools and complex topography. This aligns with Penone et al.'s (2016) finding that temperate biomes (e.g., coniferous forests) exhibit lower trait beta diversity across realms due to historical connectivity (e.g., Pleistocene dispersal corridors) and broader climatic niches of temperate species. Mazel et al. (2017) further emphasized that temperate assemblages often retain higher trait diversity due to a balance between niche conservatism and adaptive flexibility, which can mitigate the homogenizing effects of disturbance. In our TCF sites, rugged terrain creates microhabitat refugia that allow disturbance-sensitive species to persist in localized patches, leading to increased compositional heterogeneity rather than homogenization. This also reflects Rovero et al.'s (2020) observation that temperate biomes with complex topography are less prone to functional homogenization, as refugia maintain trait diversity even under moderate disturbance. Collectively, these biome-specific patterns reinforce Penone et al.'s (2016) core insight that environmental filters drive parallel assemblage structure across biomes globally, while extending this framework to show how human pressures interact with these filters to produce context-dependent responses.

Our results indicate that human modification at the regional scale imposes the strongest disturbance on Southeast Asian mammal communities: it induced marked homogenization and compositional shifts and yielded divergent functional patterns (supporting H1). Similar restructuring has been observed in river macroinvertebrate assemblages (Mouton et al. 2020). The observed functional divergence can arise through two nonexclusive mechanisms: (i) reduced availability of natural resources intensifies competitive exclusion and promotes niche partitioning (Hardin 1960); and (ii) human-modified landscapes such as agricultural mosaics create heterogeneous environments (Duelli 1997) and provide anthropogenic resources such as food

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and shelter for certain wildlife species (Hansen et al. 2020), broadening their realized trait space. Contrary to our prediction, the mammal communities in TSMBF also support H1 rather than H3. The phenomenon of increased SES.FRic driven by human modification and habitat fragmentation in TSMBF may stem from species rapidly accumulating in warm, high-productivity forests despite limited time for ecological divergence (Oliveira et al. 2016). The function of reference communities exhibits greater redundancy than in TCF communities, leading to functional divergence following landscape alteration.

Notably, human presence also promoted homogenization across all areas and within TSMBF, indicating that direct human interference and landscape alteration jointly exert a strong influence on the species pools. Southeast Asia remains a hotspot for wildlife hunting and trade (Hughes et al. 2023) with frequent human–wildlife conflict (Wierucka et al. 2023); under these conditions, fear of humans can trigger avoidance behaviors (Lee et al. 2024; Mendes et al. 2020), leading to asymmetric survival among species and benefits to disturbance-tolerant taxa (Suraci et al. 2021). Our findings align with previous work showing that, under rapid human expansion, disturbance-intolerant mammals decline, whereas a few widespread species replace endemic specialists, leading to large-scale homogenization and accelerating extinctions (McDonnell and Hahs 2015).

Building-density change similarly promoted compositional homogenization at the regional scale but showed a context-dependent association with SES.FRic. In our dataset, the building-density change ranged from 0 (indicating negligible change) to 40.08 (indicating decades of increase). Across all study areas, the functional pattern showed a hump-shaped relationship with building-density change—divergence at intermediate levels followed by convergence at high levels. In TSMBF, by contrast, the pattern declined monotonically with increasing building-density change, which was slightly higher in TSMBF than in TCF, although the difference was nonsignificant (Wilcoxon test, $p = 0.062$). Together, these results support H3 under high building-density change, consistent with previous findings for mammals and birds (Cooke, Eigenbrod, and Bates 2019; Liang et al. 2019). Moderate land-use can increase trait differences by introducing disturbance-adapted species (Kuipers et al. 2023), whereas intensive changes filter out habitat-dependent taxa, leaving trait-redundant generalists and reducing functional diversity (Boron et al. 2019; Suraci et al. 2021). Comparing building-density change with human modification underscores the need to consider time series: prolonged exposure may generate extinction debts, in which richness and functional diversity measured during relaxation do not fully reflect impending defaunation (Hylander and Ehrlén 2013).

Patch density was positively correlated with FRic, FDis, and FEve across all study areas and was significantly associated with functional divergence. These patterns are consistent with the idea that certain forms and magnitudes of habitat heterogeneity associated with fragmentation can increase biodiversity. Habitat heterogeneity promotes niche partitioning, supports more species with distinct traits, reduces competition, and facilitates coexistence (Chen et al. 2020; Mortelliti et al. 2010; Szangolies et al. 2022). Moreover, habitat edges and cores may favor different species, increasing functional diversity (Betiol et al. 2023; Haight et al. 2023; Jin et al. 2021). Nevertheless, fragmentation can strongly disadvantage forest habitat specialist species (Keinath et al. 2017). Further work is needed to identify thresholds of fragmentation that sustain biodiversity and clarify species-specific responses.

Our results show that the effects of human pressures differ by biome (supporting H4). Human pressures explained more variation in functional patterns within the TSMBF than the TCF (Table S3), and compositional homogenization and shifts were more severe in the TSMBF (Figure 3). Within the TCF, changes in building-density, human modification, and fragmentation were associated with heterogenization, and the role of fragmentation was consistent with H2. Biogeographic and compositional contrasts likely contribute. The TCF study areas are largely mountainous, where altitude and terrain roughness were strongly associated with community structure (Table S3). Complex topography and montane microclimates can increase stochasticity in post-disturbance survival, movement, and extinction (Banks et al. 2015; Lewis et al. 2021; Pierce et al. 1999), amplifying among-region compositional differences. In contrast, tropical lowland humid ecosystems, including TSMBF, are less seasonal, where wildlife evolved under narrower climatic ranges (Newbold et al. 2020) and may be more vulnerable to human-driven habitat changes. Our findings suggest that tropical homogenization stems mainly from increases in generalist species such as the wild boar (*Sus scrofa*) and masked palm civet (*Paguma larvata*) and declines in endemic/threatened species such as the banded palm civet (*Hemigalus derbyanus*) and the sun bear (*Helarctos malayanus*), which consequently reduce broad-scale diversity and risk ecosystem decay (Blandón et al. 2016; Laurance et al. 2002). With tropical forests facing ongoing exploitation and loss (Myers 1983), anthropogenic disturbance coupled with habitat loss has driven steep wildlife declines, especially among sensitive species (Greco et al. 2025; Fernanda Michalski and Peres 2007; Symes et al. 2018). We therefore call for intensified conservation in tropical lowland humid forests to preserve intact habitats and retain living space for endangered populations. This does not imply that communities in temperate biomes are secure: grazing was more frequent in temperate than in tropical sites (Wilcoxon test, $p < 0.001$) and more strongly associated with SES.FRic than human modification (Figure 5, Table S3), warranting closer scrutiny of its ecosystem impacts.

Meeting conservation goals will require ambitious strategies to safeguard habitats while reversing biodiversity declines (Leclère et al. 2020; Stephens 2023). Community assembly processes under disturbance have long been central to ecological succession theory and are especially pertinent amid Anthropocene threats (Helsen et al. 2012; Li and Shipley 2018; Sytsma et al. 2023). By integrating compositional and functional indicators, we quantified mammal community responses to multiple human pressures and highlighted the particular vulnerability of tropical regions. We recommend adaptive biome-specific management (Keith et al. 2011; Nguyen et al. 2025), including mitigating forest degradation in tropical regions to protect endemic specialists (Grantham et al. 2020; Green et al. 2020) and enforcing regulations where high-intensity human intrusion and grazing occur (Luo et al. 2025; Soofi et al. 2018). Our findings offer a complementary lens for assessing human impacts

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and informing policies aimed at achieving conservation targets and the sustainable use of biodiversity.

Author Contributions

Xueyou Li: conceptualization, methodology, software, validation, supervision, funding acquisition, writing – original draft, writing – review and editing, visualization, project administration, resources, data curation. **Xuelong Jiang:** supervision, project administration, resources. **Kenneth Otieno Onditi:** writing – review and editing, validation, visualization. **Wenqiang Hu:** data curation, investigation. **Yang Liu:** formal analysis, software, data curation, writing – original draft, writing – review and editing, methodology, investigation, validation, visualization.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data and model code for supporting the findings are openly available in ScienceDB at: https://doi.org/10.57760/sciencedb.29775.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Information on the study areas. Lon, lontitude; Lat, latitude; n_Cameras, the number of camera traps; Effort, survey effort (day). Referance represents the data origin. Spacing represents the average spacing between camera traps. For each biome where the study area is located, "TCF" stands for Temperate Conifer Forests, "TSMBF" stands for Tropical & Subtropical Moist Broadleaf Forests, and "TSDBF" stands for Tropical & Subtropical Dry Broadleaf Forests. **Table S2:** LRR indices. Covariate, the type of human pressures; Index, the type of LRR indices; LRR, the value of the LRR index. For the human pressures; PD, patch density; HM, human modification; BC, building-density change; RAIH, relative abundance index for humans; RAIL, relative abundance index for livestock. For Biome of each LRR index, "TCF" stands for Temperate Conifer Forests, "TSMBF" stands for Tropical & Subtropical Moist Broadleaf Forests, and "Total" stands for all study are as included in this study. 95% CI and 90% CI represent the 95% and 90% confidence intervals of LRR indices, respectively. **Table S3:** Results of the functional diversity GLM model. Covariate includes environmental covariates and their quadratic terms, Intercept, and offset. Index, the type of functional diversity indices; Estimate, the regression coefficient of the GLM models; Std, the standard error of the regression coefficient; EV, the explained variance (%). For the environmental covariates, PD, patch density; HM, human modification; BC, building-density change; RAIH, relative abundance index for humans; RAIL, relative abundance index for livestock; Alt, altitude; Rough, terrain roughness; Bio14, precipitation of driest month; NPP, net primary productivity; TC, tree-cover change. For Biome of each result, "TCF" stands for Temperate Conifer Forests, "TSMBF" stands for Tropical & Subtropical Moist Broadleaf Forests, and "Total" stands for all study are as included in this study. 95% CI and 90% CI represent the 95% and 90% confidence intervals of the regression coefficient, respectively. **Figure S1:** Covariate collinearity. (a) Pearson correlation coefficients of environmental covariates extracted using different buffer radii. (b) The Pearson correlation between environmental covariates extracted within a 5 km buffer radius. "×" indicates a p-value > 0.05. For the environmental covariates, PD, patch density; HM, human modification; BC, building-density change; RAIH, relative abundance index for humans; RAIL, relative abundance index for livestock; Alt, altitude; Rough, terrain roughness; Bio14, precipitation of driest month; NPP, net primary productivity; TC, tree-cover change. **Figure S2:** Null model normality for LRR homogeneity. Frequency distribution and Kolmogorov–Smirnov test results of LRR homogeneity. For the environmental covariates, PD, patch density; HM, human modification; BC, building-density change; RAIH, relative abundance index for humans; RAIL, relative abundance index for livestock; Alt, altitude; Rough, terrain roughness; Bio14, precipitation of driest month; NPP, net primary productivity; TC, tree-cover change. For Biome of each result, "TCF" stands for Temperate Conifer Forests, "TSMBF" stands for Tropical & Subtropical Moist Broadleaf Forests, and "All areas" stands for all study areas included in this study. **Figure S3:** Null model normality for LRR shift. Frequency distribution and Kolmogorov–Smirnov test results of LRR shift. For the environmental covariates, PD, patch density; HM, human modification; BC, building-density change; RAIH, relative abundance index for humans; RAIL, relative abundance index for livestock; Alt, altitude; Rough, terrain roughness; Bio14, precipitation of driest month; NPP, net primary productivity; TC, tree-cover change. For Biome of each result, "TCF" stands for Temperate Conifer Forests, "TSMBF" stands for Tropical & Subtropical Moist Broadleaf Forests, and "All areas" stands for all study areas included in this study.

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